Q 1

1. $m(a+bX) = \frac{1}{N}\sum_{i=1}^{N}(a+bx_i) = \frac{1}{N}\left(\sum_{i=1}^{N} a + \sum_{i=1}^{N} bx_i\right)$

$= \frac{1}{N}\left(Na + b\sum_{i=1}^{N} x_i\right) = a + \frac{b}{N}\sum_{i=1}^{N} x_i$

$m(X) = \frac{1}{N}\sum_{i=1}^{N} x_i,$

$\underbrace{= a + b(m(X)) = m(a+bX)}$

2. $cov(X, a+bY) = \frac{1}{N}\sum_{i=1}^{N}(x_i - m(X))(y_i - m(a+bY))$

$= \frac{1}{N}\sum_{i=1}^{N}(x_i - m(X))((\cancel{a}+by_i) - (\cancel{a}+bm(Y)))$

$= \frac{1}{N}\sum_{i=1}^{N}(x_i - m(X))(b(y_i - m(Y)))$

$cov(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - m(X))(y_i - m(Y))$

$\underbrace{= b\,cov(X,Y) = cov(X, a+bY)}$

3. $cov(a+bX, a+bX) = \frac{1}{N}\sum_{i=1}^{N}((a+bx_i) - m(a+bX))^2$

$= \frac{1}{N}\sum_{i=1}^{N}((\cancel{a}+bx_i) - (\cancel{a}+bm(X)))^2 = \frac{1}{N}\sum_{i=1}^{N} b^2(x_i - m(X))^2$

$\therefore \underbrace{= b^2 cov(X,X)} \rightarrow \dfrac{\frac{1}{N}\sum_{i=1}^{N} b^2(x_i - m(X))^2}{b^2} = s^2$ ✓

4. $g(x) = x^2 \rightarrow m(g(\{1,2\})) = \frac{1+4}{2} = \frac{5}{2}$

$g(m(\{1,2\})) = \frac{1+2}{2} = \frac{3}{2} \rightarrow \left(\frac{3}{2}\right)^2 = \frac{9}{4} \neq \frac{5}{2}$

$\therefore \underbrace{g(m(x)) \neq m(g(x))}$